

Stress-induced single-polarization single-transverse mode photonic crystal fiber with low nonlinearity

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Abstract: We report on the design of a single-polarization single-transverse mode large mode area photonic crystal fiber. By including index-matched stress applying elements in the photonic cladding an ultra-broadband single polarization window is obtained while a large mode field area of $\sim 700 \mu\text{m}^2$ is maintained. Based on that design, an Yb-doped double-clad photonic crystal fiber is realized that combines low nonlinearity and single polarization properties. A first result of the high power operation using this fiber is demonstrated.

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1. Introduction

Many experiments have proven the potential of power scaling when using rare earth doped fibers as a gain medium in lasers and amplifiers [1]. Recently, photonic crystal fibers (PCF) pushed the output parameters of solid-state laser systems not only in continuous wave operation [1,2] but especially in ultra-short pulse systems [3].

Photonic crystal fibers usually consist of a regular array of air-filled holes and a defect in its center, e.g. a missing air hole, which defines the core. The so formed photonic cladding is responsible for an extended parameter range of optical properties in these fibers, e.g., large waveguide dispersion, endlessly single mode operation and low nonlinearity [1,4]. These properties provide scaling potential for fiber laser and amplifier systems. In particular, due to the scaling of the nonlinear limits, which are greatly reduced due to the larger single mode cores and shorter absorption of air-clad large mode area photonic crystal fibers [5], a dramatically scaling of the output of ultra-short pulsed fiber laser systems is achieved. In some of these high power experiments a degradation of the degree of polarization is observed [6]. To overcome this problem and especially to simplify the laser setup in terms of polarization control there is a great interest in combining large mode area microstructured fibers and polarization maintaining elements.

For polarization maintenance sufficient birefringence has to be introduced. This can be achieved by form birefringence, e.g., elliptical cores, or material anisotropy. The implementation of form birefringence suffers from the disadvantage, that the birefringence decreases rapidly for larger cores. Thus, material anisotropy has to be introduced into the fiber core. This is usually done by use of the well-known technique of stress-applying parts (SAP) inside the fiber, where the elasto-optical effect introduces anisotropy and therefore birefringence. In addition, there is the advantage of a relatively low wavelength dependence of the stress-induced birefringence [7], which in combination with the large single-mode wavelength range of photonic crystal fibers [8] can provide a large highly birefringent

bandwidth [9]. In a recent paper we showed, that the air holes in the photonic cladding do not screen the stresses that are applied whether by an external force or by SAPs [10]. We also concluded that a low nonlinearity fiber for application in ultrafast optics is a challenging task when using SAPs outside the photonic cladding. The main reason for this is the large space between the SAPs required for the photonic cladding, thus reducing the birefringence, but also the large outer cladding comprising the SAPs. The second argument reduces the absorption length due to a low overlap of pump radiation with the doped core in this double-clad fiber. Nevertheless, such fibers have been fabricated but so far they are limited to a mode field area of about $\sim 180 \mu\text{m}^2$ and an estimated absorption length of several meters [9,11].

In this contribution we report on the design of photonic crystal fibers that comprises stress-applying elements as part of the photonic cladding. Beside the stress-induced birefringence, the light is confined by both parts of the photonic cladding: the air holes and the index matched regular array of SAPs. We show that the birefringence is enough to split two polarization states of the weakly guided fundamental mode in that way, that the effective index of one polarization is below the cladding index, thus, resulting in a single polarization large mode area fiber. The mode field area of $\sim 700 \mu\text{m}^2$ is larger than that of any step-index single mode fibers. Additionally, the value of normalized polarization bandwidth $\Delta\lambda/\lambda_c > 0.5$ is, to the best of our knowledge, the largest ever reported for fibers so far. Furthermore, the structure is improved to a low nonlinearity Yb-doped air-clad single polarization single-transverse mode photonic crystal fiber and first experimental results are presented.

2. Design of the polarization maintaining low nonlinearity photonic crystal fiber

2.1 Basic considerations

A photonic crystal fiber with a hexagonal holey cladding (here called inner cladding) is shown in Fig. 1. The guiding property in such solid core PCF is only determined the air-hole diameter d and the pitch Λ [1,4]. For $d/\Lambda < 0.45$ the one-hole missing fiber is single mode for all wavelengths [8]. Thus, for a fixed wavelength the mode field diameter can be scaled up, limited by propagation losses [12]. A further increase in the mode field diameter can be made by removing more than one air-hole, core diameters up to $\sim 50 \mu\text{m}$ have been demonstrated with seven missing air-holes [6].

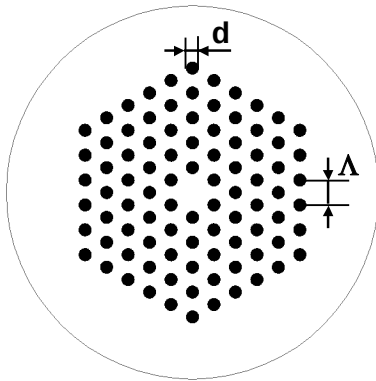


Fig. 1. Design and parameters of a micro-structured fiber: Λ – pitch, d – air-hole diameter

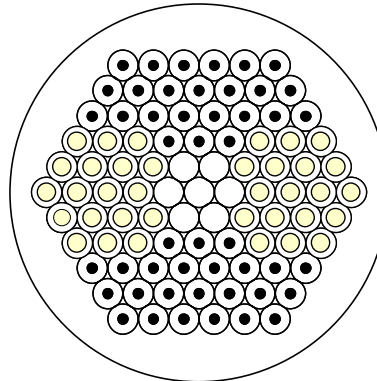


Fig. 2. Design and parameters of polarization maintaining large mode area (seven missing holes) photonic crystal fiber comprising index-matched stress applying elements (yellow) as part of the photonic cladding.

The goal of this work is to design a low nonlinearity birefringent fiber applicable to lasers and amplifiers. The requirements for low nonlinearity reduce to a large single mode core and a short length, which in this case is the absorption length. Therefore the pump cladding has to be minimized, where the limit is given by the size of the inner cladding. To achieve this, we move the stress applying elements, which are usually placed apart from the core (or even the inner photonic cladding) close to the core. At the same time, the guiding properties of the

single-mode core should not be affected. The basic idea as so to achieve this goal is shown in Fig. 2. The stress applying elements consist of a material with different thermal expansion coefficient α than the surrounding cladding material (fused silica (FS)). Using Boron-doped silica (BS) with $\alpha_{BS}=5\cdot 10^{-7}/K$ compared to $\alpha_{FS}=10\cdot 10^{-7}/K$ a permanent stress field can be generated when cooling the fiber below the softening temperature during the drawing process. Owing to the fact that the refractive index of this material is lower than that of fused silica ($\Delta n=-0.008$) a similar periodic inner cladding compared to that of the air-hole cladding can be constructed (see Fig. 2) by matching the effective cladding indices to be equal. This index matching, the arrangement of the SAPs and the achievable birefringence are discussed in the next paragraph.

2.2 Theoretical evaluation of the new design

Because of the time consuming and complex theoretical modeling of the design caused by a combined stress and waveguide analysis, our evaluation of the fiber design is done in two steps. Firstly, the boron-doped stress applying regions are index matched to the air-hole cladding, which also ensures that the effective indices of the core mode are matched [13]. As a result, both claddings can, to first order, be arbitrary mixed in a way, that by the arrangement of the SAPs a desired (e.g., maximum) birefringence is created. The arrangement and number of SAPs replacing the air holes is the second design step towards the final design.

To match the effective cladding indices, the indices of the first space-filling mode have to be equal. This calculation is carried out numerically by a plane wave expansion method using a freely available software package [14]. Simple geometric considerations fail because for large mode area fiber, the pitch is much larger than the wavelength ($\lambda \ll \Lambda$) so that the effective index is close to that of fused silica. The values for this calculation, where the matching is achieved, are the following: For the air-hole cladding we used the parameters of our previous non-polarization maintaining fiber, which was strictly single mode with $\Lambda=12.3\ \mu\text{m}$ and $d_{\text{air-hole}}/\Lambda=0.09$ [6], shown in Fig. 3. The matching of the cladding indices resulted in values of the fused silica surrounded boron-doped elements of $d_{BS}/\Lambda=0.25$ assuming a refractive index shift of this material of $\Delta n=n_{FS}-n_{BS}=-0.008$ (Fig. 4). The refractive index of fused silica is $n_{FS}=1.4422$ and the wavelength is set to $\lambda=1064\ \text{nm}$. With these parameters both photonic claddings provide single mode operation, where the fundamental mode has an effective index of $n_{\text{eff}}=1.44213$.

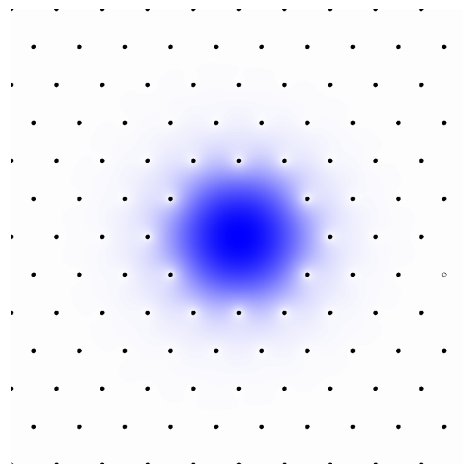


Fig. 3. Inner cladding formed by air-holes providing a large single mode core.

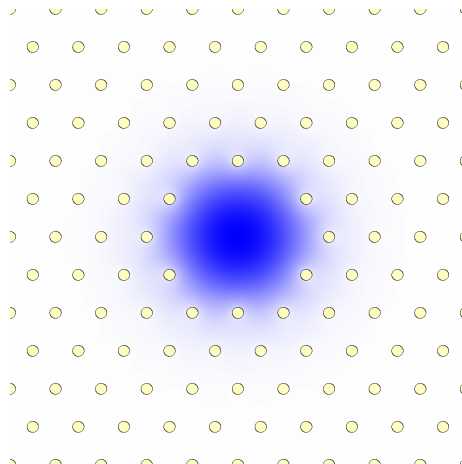


Fig. 4. Inner cladding formed by Boron-doped silica rods providing a large single mode core, which is matched to that of Fig. 3.

The stresses resulting from the different thermal expansion of the materials can be calculated according to the equilibrium equation (Eq. (1)), where σ is the stress tensor, $\epsilon_{x/y}$ are the normal strain components, γ_{xy} is the shear strain component, $\mathbf{D}$ is the elasticity matrix describing an isotropic material using Young's modulus E and Poisson's ratio ν , α is the expansion coefficient, T the high temperature and T_{ref} the reference temperature, e.g. room temperature [15].

$$-\nabla \sigma = -\nabla \cdot \begin{pmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{pmatrix} - \begin{pmatrix} \alpha \\ \alpha \\ 0 \end{pmatrix} (1 + \nu)(T - T_{ref}) = 0 \quad \text{where } \sigma = \mathbf{D}\epsilon \quad (1)$$

Using the photo-elastic tensor for fused silica, which is taken as the core material, the refractive index distribution resulting from stress/strain fields can be calculated. According to the elasto-optical effect $\Delta n = -C\sigma$ the bulk birefringence B in the center and the average value B_{av} over a certain core radius is evaluated by Eq. (2) using $C_1 = 4.2 \cdot 10^{-12}/\text{Pa}$ and $C_2 = 0.65 \cdot 10^{-12}/\text{Pa}$. It has been shown, that the core and average birefringence are an almost equivalent figure of merit for the achievable modal birefringence even for small core photonic crystal fibers [10,16].

$$B = |(C_2 - C_1)(\sigma_x - \sigma_y)| \quad (2)$$

$$B_{av} = |(C_2 - C_1) \iint (\sigma_x - \sigma_y) r dr d\varphi|$$

The problem for the complex geometries of photonic crystal fiber is solved numerically using finite element method (FEM). The parameters taken are the following: $T = 1000^\circ\text{C}$, $T_{ref} = 20^\circ\text{C}$, $\alpha_{BS} = 5 \cdot 10^{-7}/\text{K}$, $\alpha_{FS} = 10 \cdot 10^{-7}/\text{K}$, $E_{FS} = E_{BS} = 72 \cdot 10^9 \text{ Pa}$ and $\nu = 0.17$.

Four different fibers, shown in Fig. 5(b)-(e), have been compared theoretically in terms of achievable birefringence. Firstly, a large-mode-area step index PANDA-type fiber with an outer diameter of 400 μm and a core of 30 μm is compared with a similar fiber including an inner photonic cladding to confine the light in the core (Fig. 5(b),(c)). The position and size of the PANDA stress rods are evaluated using analytical expressions [17]. The stress-rods of the first fiber have a diameter of $D_{SR} = 130 \mu\text{m}$ and a center-to-center distance of $x_{cc} = 210 \mu\text{m}$. For the fiber with inner photonic cladding the optimized values are $D_{SR} = 106 \mu\text{m}$ and $x_{cc} = 238 \mu\text{m}$. A simple guideline to achieve the highest birefringence in the center of the core can be illustrated by dividing the fiber into four pieces as done in Fig. 5(a). The stress applying elements are then placed into two opposed parts to maximize the core birefringence. This guideline applies for bow-tie as well as for PANDA type fibers [18] and is used for the arrangement of the index-matched SAP in our new design. There, a number of 48 SAPs (Fig. 5(d)) and 36 SAPs (Fig. 5(e)) are arranged for an angle embedding the SAPs of 120° and 60° , respectively. The outer diameter of the new design is reduced to 170 μm .

Using the results of the FEM calculations a first conclusion about the achievable birefringence can be made: The birefringence of the step index PANDA type fiber (b) is reduced in the PANDA type PCF (c) from $B_{av} = 8.9 \cdot 10^{-5}$ to $B_{av} = 4.1 \cdot 10^{-5}$. Such fibers have been successfully demonstrated [7,9] but due to the large outer diameter representing a pump cladding in actively doped fibers and a mode field diameter of $\sim 15 \mu\text{m}$, a high nonlinearity can be predicted. In contrast, with our new design a birefringence of $B_{av} = 4.6 \cdot 10^{-5}$ is introduced which is almost half the value obtained by the PANDA step index fiber even if the out diameter is reduced to 170 μm . The calculation also showed that there is no significant difference comparing the designs of Fig. 5(d) and 5(e). If this birefringence $B = |n_x - n_y|$ is in the same order of magnitude as the confinement of the guided mode $\Delta n_m = n_{\text{fundamental_mode}} - n_{\text{cladding}}$, the fiber can show single polarization behavior [9,19]. Consequently, considering single polarization guidance, the weak guiding in the large single mode core of the PCF would require less birefringence than a similar slightly multi-mode core.

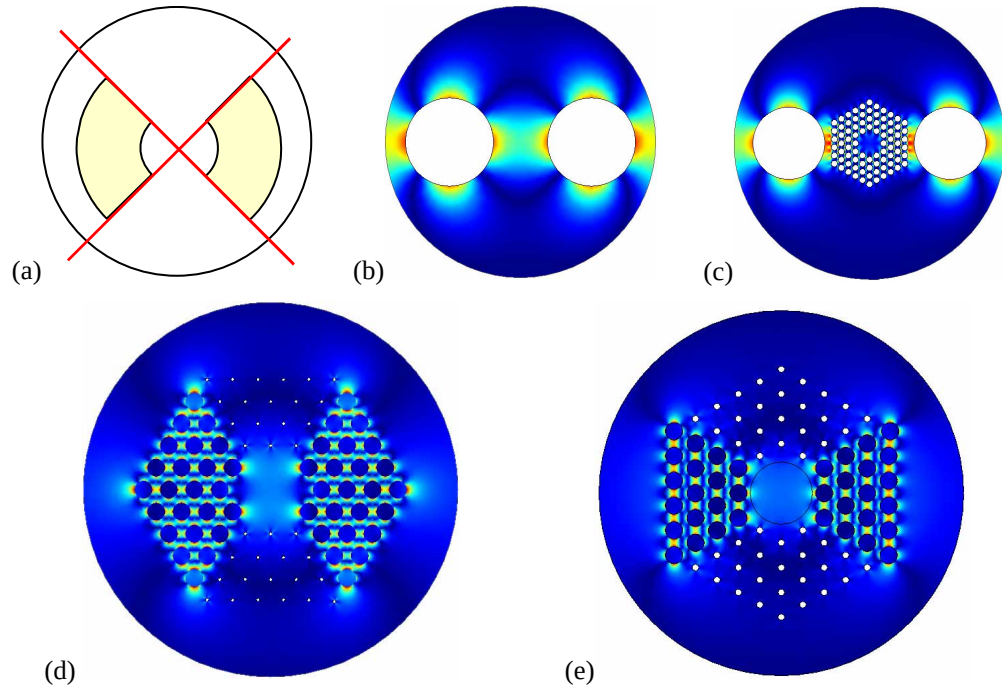


Fig. 5. (a) Illustration of the guidelines for maximizing stress induced birefringence in PANDA type fiber. (b-e): Calculated stress induced birefringence B: fiber with $D=400\text{ }\mu\text{m}$ and step index core (b), $D=400\text{ }\mu\text{m}$ but photonic inner cladding (c), fiber design with index matched stress applying parts and $D=170\text{ }\mu\text{m}$ (d,e), see text for details.

The effect of single polarization behavior in view of stress-induced birefringence should be described in more detail: If a stress field is induced by the stress-applying elements, it is translated into a change of the refractive index thanks to the photo-elastic effect. For a stress field, the index change in direction of the SAPs is positive whereas in the transverse direction the stress is reversed. This translates into a downshift of the refractive index in this direction. If the stress is high enough, this index can be shifted below the cladding index so that the polarization state in this direction is not guided anymore – the fiber becomes single polarization [19,20]. As mentioned, due to the weak guiding of the fundamental mode in a large mode area photonic crystal fiber, a relatively low stress field is necessary. From the values predicted in our simulations, we can be confident to achieve at least a polarization maintaining PCF.

An important fact of our design should be noted: Due to the necessary periodic cladding of the SAPs, a gap is introduced between the boron-doped material. This greatly reduces the stress that is usually induced in the core – the main stress is achieved between the rods, which is also indicated by the birefringence calculations in Fig. 6. There, the dependence of the birefringence on the relative size of the SAPs d_{BS}/Λ for the design of Fig. 5(d) is shown. Additionally, the influence of the number of SAPs is investigated for $d_{BS}/\Lambda=0.82$ as shown in Fig. 7 with the result, that more than half of the birefringence is already achieved when only using 6 SAPs instead of the whole 48 ones. These calculations will be used to guide the further design steps. The theoretical design is confirmed by the experimental results of the actual fibers that have been drawn, which are discussed in the next section.

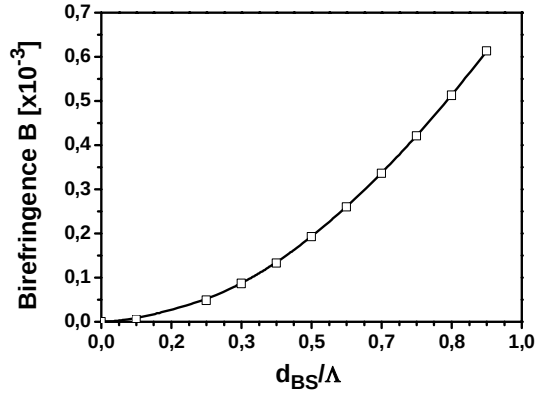


Fig. 6. Effect of changing the relative hole size of the boron-doped SAPs on the achievable birefringence.

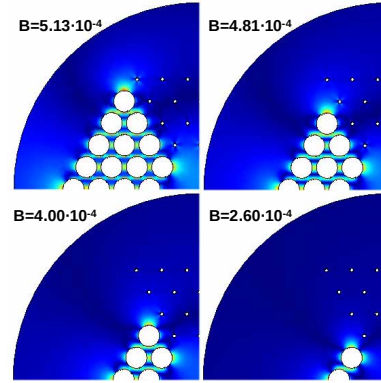


Fig. 7. Effect of removing rows of SAPs on the achievable birefringence. The color scaling is the same as Fig. 5.

3. Characterization and experimental results

3.1 Characterization of the passive reference fibers

Fibers with the design discussed in the previous section were fabricated and the results of the passive fibers are shown in Fig. 8 and Fig. 9. The cladding material is fused silica and the stress applying elements are made of boron-doped silica. Two different fibers have been fabricated, the first comprises 20 SAPs (PassivePCF01) and the second one only 6 SAPs (PassivePCF02). Reducing the number of SAPs simplifies the fabrication of the fiber as less overall stresses are introduced. Nevertheless, most of the birefringence is introduced by the inner SAPs as shown in the previous paragraph (Fig. 7).

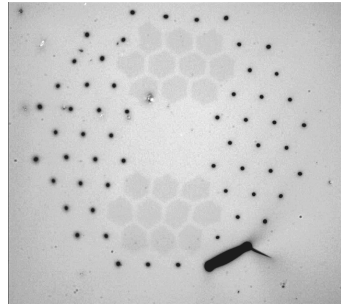


Fig. 8. Microscope images of the PCF (PassivePCF01) where 20 holes are replaced by index-matched stress applying parts

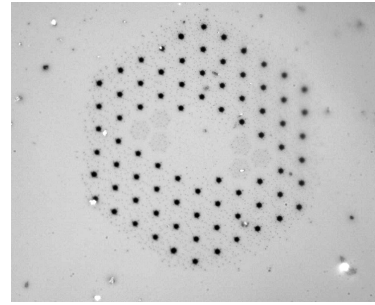


Fig. 9. Microscope images of the PCF (PassivePCF02) where 6 holes are replaced by index-matched stress applying parts

The parameters for PassivePCF01 are $\Lambda=12.3 \mu\text{m}$, $d_{\text{air-hole}}/\Lambda=0.12$ and $d_{\text{BS}}/\Lambda=0.85$. Compared to the design guidelines in the previous chapter, the values indicate that the fiber will not match the single-mode condition. This is in fact true, but the fiber becomes single-mode at a bending radius of $R < 50 \text{ cm}$. Figure 10 shows the transmission properties for the two polarization states for PassivePCF01 at bending diameters of 1.4 m, 0.25 m and 0.15 m, respectively. The fiber length is 1 m. It can be clearly seen that the fiber exhibits a certain bandwidth where only one polarization is guided. This bandwidth, defined as a 10 dB drop in the transmission, ranges from $\sim 750 \text{ nm}$ up to 1250 nm for $R=1.4 \text{ m}$ and is still very large in the single mode case of $R=0.25 \text{ m}$ where it ranges from 880 nm above 1600 nm , limited by the range of the spectrometer that was used. The normalized bandwidth $\Delta\lambda/\lambda_c > 0.5$ of that polarizing window is the highest value ever reported for fibers [9] in combination with a

measured mode field area of $\sim 700 \mu\text{m}^2$ for the fundamental mode. It can also be observed, that the short-wavelength limit shifts due to macro bending losses [21].

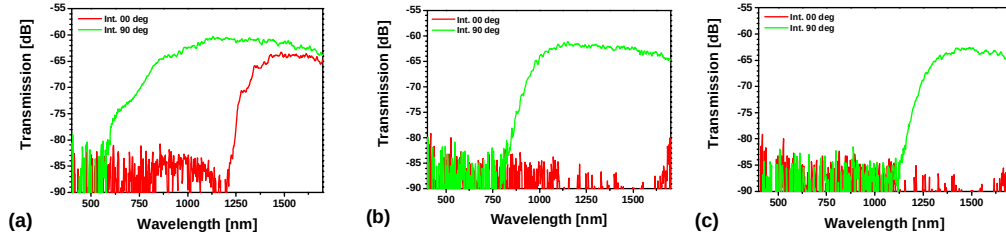


Fig. 10. Transmission spectra for the two polarization states using the fiber PassivePCF01 and different bending diameters of (a) 1.4 m, (b) 0.25 m and (c) 0.15 m.

The second fiber (PassivePCF02) incorporates less SAPs and has the relative hole size of $d_{\text{air-hole}}/\Lambda=0.2$, $\Lambda=12.3 \mu\text{m}$ and $d_{\text{BS}}/\Lambda=0.85$. The less SAPs the less birefringence is obtained as discussed in the previous section. This translates into a smaller polarizing (PZ) window as can be see from the measurement at a bending radius $R=50 \text{ cm}$ and a fiber length of 2.5 m shown in Fig. 11(a). At higher wavelengths, the fiber is birefringent with the values display in Fig. 11(b). The dependence of the PZ window on the bending has been determined in detail for larger relative air holes sizes $d_{\text{air-hole}}/\Lambda \sim 0.26 \dots 0.30$ in order to examine the polarization guiding properties at tighter bending. The result is shown in Fig. 12. Reducing the bending diameter shifts the PZ windows to higher wavelength, but the bandwidth of 140 nm is maintained. Clearly, increasing $d_{\text{air-hole}}/\Lambda$ lead to a downshift of the PZ window due to the increased guiding strength translating into less macro bending losses.

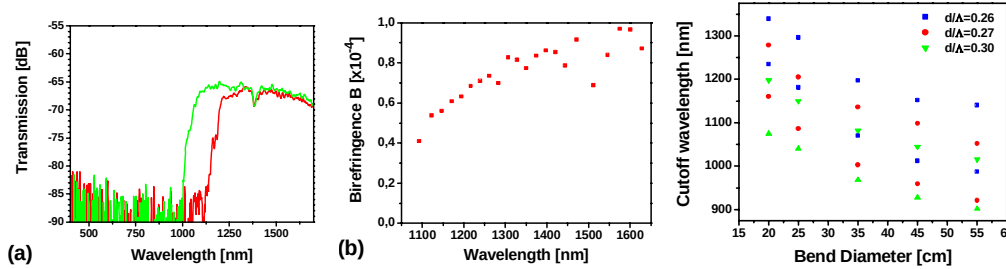


Fig. 11. Transmission spectrum (a) and birefringence (b) measured in the fiber PassivePCF02 with $d/\Lambda=0.2$, a bending diameter of 0.5 m and a length of 2.5 m.

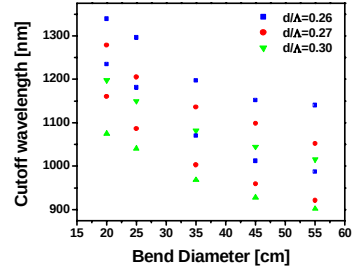


Fig. 12. Polarizing window dependence on $d_{\text{air-core}}/\Lambda$ and the bending diameter.

All the characterizations have been done at room temperature. It is clear that SAP induce a temperature dependent birefringence. In a future work, we will characterize the thermal sensitivity of these fibers, which should be similar to thermal sensitivity of the birefringence in standard PANDA type fibers. To first order, the birefringence varies linearly with temperature [18,22]. Thus, at reasonable high temperatures, changes in the single-polarization window are expected.

3.2 Design and experimental results of the Yb-doped single-polarization fiber

The PZ window of fiber PassivePCF02 shown in Fig. 11 lies within the emission bandwidth of ytterbium. Therefore we used this structure to fabricate an Yb-doped double clad fiber. The inner photonic cladding is surrounded by an air-clad structure to guide the pump light. The structure is comparable to the recently studied non-polarization maintaining low nonlinearity photonic crystal fiber [6]. An image of the fabricated fiber is shown in Fig. 13. The pump cladding has a diameter of $170 \mu\text{m}$ and a NA of ~ 0.62 . The inner cladding is the same as the passive reference fiber described before forming a core with a diameter of $35 \mu\text{m}$. The numerical aperture of this core is $\text{NA} \sim 0.03$ providing almost single transverse mode at straight with a measured mode field diameter of $27 \mu\text{m} \times 33 \mu\text{m}$. The slightly elliptical mode results

from the mismatch of the cladding indices but of course, this will not give rise to additional form birefringence at this large core and low NA. The different single mode behavior is in contrast to the passive reference fiber described before, which had to be bent. One reason for this is that the inner core is matched to the larger values of d/Λ , meaning the average refractive index of the fluorine down-doped Yb-core is decreased below the refractive index of fused silica. This ensures almost single-mode behavior even though the parameters of the cladding are not perfectly matched to the values described in section 2. Beside this uncertainty in the refractive index of the Yb-doped core, the index of the boron doped SAPs can differ from those values used in the theoretical model.

Nevertheless, the fiber was used in a first laser experiment to prove the polarizing property. Therefore a simple free running laser was built without any polarizing element inside the cavity. The setup is shown in Fig. 14. The fiber is pumped by a pigtailed laser diode. The cavity is built by one high reflection mirror on the one side and the 4% Fresnel reflection from the other fiber side, which has been perpendicularly polished. The output is characterized by a thin-film polarizer using the definition of the degree of polarization $DOP = |P_1 - P_2| / (P_1 + P_2)$ where P_1 is the maximum and P_2 the minimum power obtained by rotation the polarizer.

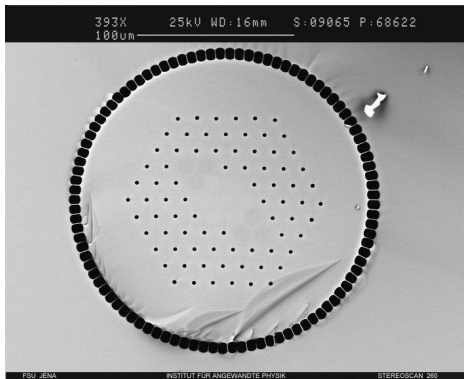


Fig. 13. SEM image of the Yb-doped air-clad photonic crystal fiber with six index-matched stress applying parts.



Fig. 14. Simple setup for characterizing the polarization properties of the fibers.

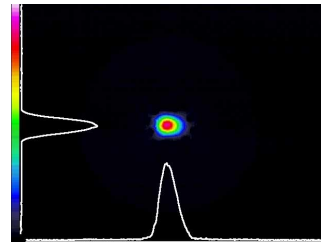


Fig. 15. Output imaged onto a CCD showing an almost Gaussian mode profile.

Two fibers have been tested by this setup. Firstly, the already mentioned non-polarization maintaining fiber with a similar photonic structure compared to our new design [6]. A fiber length of 1.2 m is used and the result is shown in Fig. 16. The slope efficiency is as high as 74% with respect to the launched pump power. The degree of polarization is below 20% for the higher power levels. It has to be mentioned that in this measurement care has to be taken to align the high reflection mirror. Any misalignment in the angle will favor one polarization state in the laser and fudge the measurement. Thus, the mirror is aligned to optimum output power and cladding mode free output and is not touched when changing the power. Also, the experiment is repeated to ensure the quality of the results. Secondly, this fiber is compared with the new polarizing Yb-doped fiber. Due to the similar structure, this fiber also showed a pump light absorption of >14 dB/m at a wavelength of 976 nm. Again a length 1.2 m of the fiber is used. The result is shown in Fig. 17. The slope efficiency is measured to be 75%. The degree of polarization stays constant for all output powers at a high value of more than 94% translating in a polarization extinction ratio of almost 15.5 dB. The emission wavelength is 1040 nm. This low wavelength together with the high slope efficiency and low threshold proves the low loss in this fiber. If the polarizing window would have been shifted to longer wavelength, a shift in the laser wavelength or at least a drop in efficiency should have been occurred. Tilting the high reflection mirror slightly does not change the polarization state of

the laser output but introduces cladding mode. Fig. 15 shows the output imaged onto a CCD camera indicating that the output is a nearly diffraction-limited single-mode beam with an almost Gaussian shape.

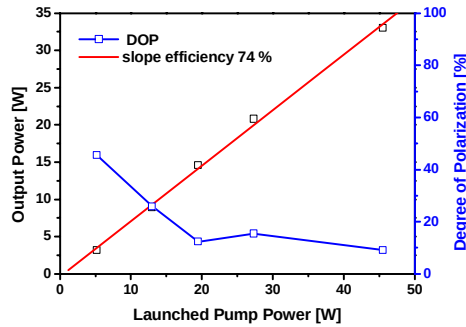


Fig. 16. Output characteristic of the non-polarization maintaining reference fiber.

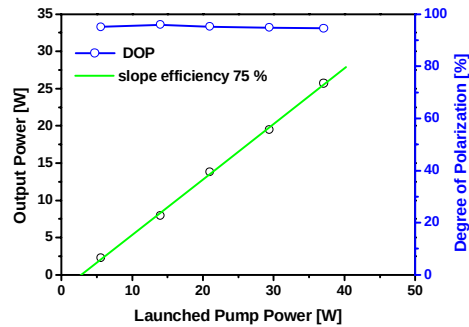


Fig. 17. Output characteristic of the single-polarization photonic crystal fiber.

4. Conclusions and outlook

In conclusion, we have shown a new design of a single-polarization single-transverse mode photonic crystal fiber. The design is based on combining stress applying elements and the inner photonic cladding by index matching for the first time. In this way, a passive single-polarization fiber with a mode field area of $\sim 700 \mu\text{m}^2$ exceeding values that can be obtained from step index fibers. Additionally, the largest single polarization bandwidth of $\Delta\lambda/\lambda_c > 0.5$ ever reported could be observed. This design was then used to fabricate an Yb-doped single polarization fiber with low nonlinearity. Because no stress elements are placed outside the inner cladding, a high overlap of the pump cladding with the large mode core could be obtained resulting in a pump light absorption of $\sim 14 \text{ dB/m}$. In a primary experiment, a highly polarized laser with an extinction ratio of 15.5 dB and an output power up to 25 W is demonstrated. In the near future, the design will be optimized and the fiber will be applied in ultra-short pulse fiber amplification systems.

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